

MSC2050 Discrete Mathematics, Week 10

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Recurrence relations

(from Paragraphs 8.1-8.3)

Fast multiplication algorithm (p.528)

Recall that multiplying two binary numbers takes $O(n^2)$ time.
Sum of two binary numbers takes $O(n)$.

The **Karatsuba algorithm** is a fast multiplication algorithm.

It was discovered by Anatoly Karatsuba in 1960 and published in 1962.

It reduces the multiplication of two n -digit numbers to $O(n^{1.58})$.

For example, the Karatsuba algorithm requires 59,049 single-digit multiplications to multiply two 1024-digit numbers, whereas the classical algorithm requires 1,048,576 (a speedup of 17.75 times).

Let $a = (a_{2n-1} \dots a_1 a_0)_2$ and $b = (b_{2n-1} \dots b_1 b_0)_2$.

Denote

$$A_1 = (a_{2n-1} \dots a_n)_2, \quad A_0 = (a_{n-1} \dots a_0)_2$$

$$B_1 = (b_{2n-1} \dots b_n)_2, \quad B_0 = (b_{n-1} \dots b_0)_2.$$

So

$$a = 2^n A_1 + A_0, \quad b = 2^n B_1 + B_0.$$

Hence,

$$ab = (2^{2n} + 2^n)A_1B_1 + 2^n(A_1 - A_0)(B_0 - B_1) + (2^n + 1)A_0B_0.$$

So the time complexity is

$$T(2n) = 3T(n) + Cn$$

or

$$T(n) = 3T(n/2) + Cn.$$

By Master theorem with $a = 3$, $b = 2$ and $d = 1$, so $3 > 2^1$,
hence $O(n^{\log_2 3 \approx 1.58})$.

Answer Q1

Fibonacci numbers

The **Fibonacci numbers**, $f_0, f_1, f_2, \dots$, are defined by $f_0 = 0$, $f_1 = 1$, and

$$f_n = f_{n-1} + f_{n-2}, \quad n = 2, 3, 4, \dots$$

The sequence does have lots of other interesting and practical applications.

Please read more about it [here](#).

The original problem that Fibonacci investigated (in the year 1202) was about how fast rabbits could breed in ideal circumstances.

Suppose a newly-born pair of rabbits, one male, one female, are put in a field. At the end of its second month a female can produce another pair of rabbits. Suppose that our rabbits never die and that the female always produces one new pair (one male, one female) every month from the second month on. The puzzle that Fibonacci posed was How many pairs will there be in one year?

Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	Total pairs
		1	1
		2	1
		3	2
		4	3
		5	5
		6	8

The number of pairs of rabbits in the field at the start of each month is 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Answer Q2

Algorithms of finding Fibonacci Numbers

Recursive Algorithm

```
procedure fibonacci( $n$ : nonnegative integer)
if  $n = 0$  then return 1
else if  $n = 1$  then return 1
else return fibonacci( $n - 1$ ) + fibonacci( $n - 2$ )
```

Number of additions is $f_n - 1$, this is $O(2^n)$.

Iterative Algorithm

```
procedure iterative fibonacci( $n$  : nonnegative integer)
if  $n = 0$  then return 1
else  $x := 1, y := 1$ 
    for  $i := 1$  to  $n - 1$ 
         $z := x + y$ 
         $x := y$ 
         $y := z$ 
    return  $y$ 
```

Number of additions is $O(n)$.

Show that whenever $n \geq 3$,

$$f_n > \left(\frac{1 + \sqrt{5}}{2} \right)^{n-2}.$$

Solution: Let $\alpha = \frac{1+\sqrt{5}}{2}$ and let $P(n)$ be the statement $f_n > \alpha^{n-2}$.

BASIS STEP: Since $\alpha < 2 = f_3$, so $P(3)$ is true.

INDUCTIVE STEP: Assume that $f_j > \alpha^{j-2}$, for all integers j with $3 \leq j \leq k$.

Because α is a solution of $\alpha^2 - \alpha - 1 = 0$, it follows that $\alpha^2 = \alpha + 1$.

Therefore,

$$\alpha^{k-1} = \alpha^2 \alpha^{k-3} = (\alpha + 1) \alpha^{k-3} = \alpha \cdot \alpha^{k-3} + 1 \cdot \alpha^{k-3} = \alpha^{k-2} + \alpha^{k-3}.$$

By the inductive hypothesis, we have $f_{k-1} > \alpha^{k-3}$, and $f_k > \alpha^{k-2}$.

Therefore, it follows that

$$f_{k+1} = f_k + f_{k-1} > \alpha^{k-2} + \alpha^{k-3} = \alpha^{k-1}.$$

Hence, $P(k+1)$ is true. This completes the proof by the strong induction.

Euclidean algorithm uses $O(\log b)$

Lamé's theorem

Let a and b be positive integers with $a \geq b$. Then the number of divisions used by the Euclidean algorithm to find $\gcd(a, b)$ is less than or equal to five times the number of decimal digits in b .

Proof: Recall that when the Euclidean algorithm is applied to find $\gcd(a, b)$ with $a \geq b$ this sequence of equations (where $a = r_0$ and $b = r_1$) is obtained.

$$r_0 = r_1 q_1 + r_2, \quad 0 \leq r_2 < r_1,$$

$$r_1 = r_2 q_2 + r_3, \quad 0 \leq r_3 < r_2,$$

$\vdots$

$$r_{n-2} = r_{n-1} q_{n-1} + r_n, \quad 0 \leq r_n < r_{n-1},$$

$$r_{n-1} = r_n q_n.$$

Here n divisions have been used to find $r_n = \gcd(a, b)$.

Note that the quotients $q_1, q_2, \dots, q_{n-1}$ are all at least 1.

Moreover, $q_n \geq 2$, because $r_n < r_{n-1}$.

This implies that

$$r_n \geq 1 = f_2,$$

$$r_{n-1} \geq 2r_n \geq 2f_2 = f_3,$$

$$r_{n-2} \geq r_{n-1} + r_n \geq f_3 + f_2 = f_4,$$

$\vdots$

$$r_2 \geq r_3 + r_4 \geq f_{n-1} + f_{n-2} = f_n,$$

$$b = r_1 \geq r_2 + r_3 \geq f_n + f_{n-1} = f_{n+1}.$$

It follows that $b \geq f_{n+1}$.

We have proved that $f_n > \alpha^{n-2}$, where $\alpha = \frac{1+\sqrt{5}}{2}$.

Therefore, it follows that $b > \alpha^{n-1}$.

Furthermore, because $\log_{10} \alpha \approx 0.208 > 1/5$, we see that

$$\log_{10} b > (n-1) \log_{10} \alpha > \frac{(n-1)}{5}.$$

Hence, $n-1 < 5 \cdot \log_{10} b$.

The number of divisions required to find $\gcd(a, b)$ with $a > b$ is $O(\log b)$.

Solving linear recurrence relations

(Paragraph 8.2)

Linear homogeneous recurrence relations

A **linear homogeneous recurrence relation of degree k with constant coefficients** is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

A consequence of the principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is **uniquely determined** by this recurrence relation and the k initial conditions

$$a_0 = C_0, \quad a_1 = C_1, \dots, \quad a_{k-1} = C_{k-1}.$$

$P_n = 1.11 \cdot P_{n-1}$ is a linear homogeneous recurrence relation of degree one.

$f_n = f_{n-1} + f_{n-2}$ is a linear homogeneous recurrence relation of degree two.

$a_n = a_{n-5}$ is a linear homogeneous recurrence relation of degree five.

$a_n = a_{n-1} + a_{n-2}^2$ is not linear.

$H_n = 2H_{n-1} + 1$ is not homogeneous.

$B_n = nB_{n-1}$ does not have constant coefficients.

Answer Q3

Theorem 1

Suppose that $x^2 - c_1x - c_2 = 0$ has two distinct roots x_1 and x_2 .

Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if

$$a_n = \alpha_1 x_1^n + \alpha_2 x_2^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants. The equation $x^2 - c_1x - c_2 = 0$ is called the **characteristic equation** and x_1 and x_2 are **characteristic roots**

Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$.

Solution: The characteristic equation is $x^2 - x - 2 = 0$. Its roots are $x_1 = 2$ and $x_2 = -1$. Hence, the solution is

$$a_n = \alpha_1 2^n + \alpha_2 (-1)^n,$$

From the initial conditions, $a_0 = 2 = \alpha_1 + \alpha_2$, $a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1)$.

Solving these two equations shows that $\alpha_1 = 3$ and $\alpha_2 = -1$.

The solution is $a_n = 3 \cdot 2^n - (-1)^n$.

Fibonacci numbers

Recall that the sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ and also satisfies the initial conditions $f_0 = 0$ and $f_1 = 1$. The roots of the characteristic equation $x^2 - x - 1 = 0$ are

$$x_1 = \frac{1 + \sqrt{5}}{2}, \quad x_2 = \frac{1 - \sqrt{5}}{2}.$$

By Theorem 1 we have

$$f_n = \alpha_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

The initial conditions are used to find the constants

$$\alpha_1 = \frac{1}{\sqrt{5}}, \quad \alpha_2 = -\frac{1}{\sqrt{5}}.$$

Consequently, the Fibonacci numbers are given by

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

Theorem 2

Suppose that $x^2 - c_1x - c_2 = 0$ has **only one root** x_0 .

Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if

$$a_n = \alpha_1 x_0^n + \alpha_2 n x_0^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$.

Solution: The characteristic equation is $x^2 - 6x + 9 = 0$. Its root is $x_0 = 3$.

Hence, the solution is

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n,$$

From the initial conditions, $a_0 = 1 = \alpha_1$, $a_1 = 6 = 3\alpha_1 + 3\alpha_2$.

Solving these two equations shows that $\alpha_1 = 1$ and $\alpha_2 = 1$.

The solution is $a_n = 3^n + n3^n$.

Answer Q4

Theorem 3

Suppose that the characteristic equation

$$x^k - c_1x^{k-1} - \dots - c_k = 0$$

has k distinct roots $x_1, x_2, \dots, x_k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}$ if and only if

$$a_n = \alpha_1x_1^n + \alpha_2x_2^n + \dots + \alpha_kx_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example

Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

Solution: The characteristic polynomial is $r^3 - 6r^2 + 11r - 6$. The characteristic roots are $r = 1$, $r = 2$, and $r = 3$.

Hence, the solutions to this recurrence relation are of the form

$$a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n.$$

The initial conditions give

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3,$$

$$a_1 = 5 = \alpha_1 + 2\alpha_2 + 3\alpha_3,$$

$$a_2 = 15 = \alpha_1 + 4\alpha_2 + 9\alpha_3.$$

From this we find that $\alpha_1 = 1$, $\alpha_2 = 1$, and $\alpha_3 = 2$. Hence, the unique solution is

$$a_n = 1 - 2^n + 2 \cdot 3^n.$$

Theorem 4

Suppose that the characteristic equation

$$x^k - c_1x^{k-1} - \dots - c_k = 0$$

has t distinct roots $x_1, x_2, \dots, x_t$ with multiplicities $m_1, m_2, \dots, m_t$, so that $m_1 + m_2 + \dots + m_t = k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}$ if and only if

$$\begin{aligned} a_n = & (\alpha_{10} + \alpha_{11}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})x_1^n \\ & + (\alpha_{20} + \alpha_{21}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})x_2^n \\ & + \dots + (\alpha_{t0} + \alpha_{t1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})x_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where α_{ij} are constants.

Example

Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

Solution: By Theorem 4, the general form of the solution is

$$(\alpha_{10} + \alpha_{11}n + \alpha_{12}n^2)2^n + (\alpha_{20} + \alpha_{21}n)5^n + \alpha_{30}9^n.$$

Linear non-homogeneous recurrence relations

A **linear non-homogeneous recurrence relation degree k with constant coefficients** is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

A consequence of the principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is **uniquely determined** by this recurrence relation and the k initial conditions

$$a_0 = C_0, \quad a_1 = C_1, \dots, \quad a_{k-1} = C_{k-1}.$$

$a_n = a_{n-1} + 2^n$ is a linear non-homogeneous recurrence relation of degree one.
 $f_n = f_{n-1} + f_{n-2} + n^2 + n + 1$ is a linear non-homogeneous recurrence relation of degree two.
 $a_n = a_{n-5} + n!$ is a linear homogeneous recurrence relation of degree five.

Theorem 5

If $\{a_n^{(p)}\}$ is a **particular solution** of the non-homogeneous linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}.$$

Example

Solve the recurrence relation $a_n = 3a_{n-1} + 2n$ with $a_0 = 3$.

Solution: Solution of the homogeneous relation $a_n = 3a_{n-1}$ is $a_n^{(h)} = \alpha \cdot 3^n$.
We will find a particular solution in the form $cn + d$.

We have

$$cn + d = 3(c(n-1) + d) + 2n.$$

So $c = -1$ and $d = -3/2$.

Consequently, $a_n^{(p)} = -n - 3/2$. Hence,

$$a_n = a_n^{(p)} + a_n^{(h)} = \alpha 3^n - n - 3/2.$$

Finding $\alpha = 11/6$ using the initial condition, we see that

$$a_n = 11/6 \cdot 3^n - n - 3/2.$$

Particular solutions theorem

Theorem 6

Suppose that $\{a_n\}$ satisfies the linear non-homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

and

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \dots + b_1 n + b_0) s^n,$$

where $b_0, b_1, \dots, b_t$ and s are real numbers.

When s is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form

$$(p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n$$

When s is a root of this characteristic equation and its multiplicity is m , there is a particular solution of the form

$$n^m (p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n.$$

Examples

What form does a particular solution of the linear nonhomogeneous recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} + F(n)$$

have when $F(n) = 3^n$, $F(n) = n3^n$, $F(n) = n^2 2^n$, and $F(n) = (n^2 + 1)3^n$?

Solution: The associated linear homogeneous recurrence relation is $a_n = 6a_{n-1} - 9a_{n-2}$. Its characteristic equation, $x^2 - 6x + 9 = 0$, has a single root, 3, of multiplicity two.

To apply Theorem 6,

If $F(n) = 3^n$, then $a_n = p_0 n^2 3^n$.

If $F(n) = n3^n$, then $a_n = n^2(p_1 n + p_0)3^n$.

If $F(n) = n^2 2^n$, then $a_n = (p_2 n^2 + p_1 n + p_0)2^n$.

If $F(n) = (n^2 + 1)3^n$, then $a_n = n^2(p_2 n^2 + p_1 n + p_0)3^n$.

Answer Q5

Homework:

- Read and understand EXAMPLE 3 at p. 505.
- Solve exercises # 7,11,13,19,21 at p. 510-511.
- Solve exercises # 1,3,9,13,15,19,21,23,27,29,31,33,37,45,47 at p. 524-526.
- Solve exercises # 1,3,7,9,12,13,17,19,21,35,37 at p. 535-536.